Alex Wang

github.com/alex-wang55 | alex-wang55.github.io | LinkedIn | awang.ottawa@gmail.com | +1 (343) 573-8677

Education

McGill University Expected May 2028
Bachelor of Science in Computer Science *Montreal, QC*

- Coursework: Data Structures and Algorithms, Software Design, Software Systems, Computer Systems

Skills

Languages: Python, C/C++, TypeScript, JavaScript, Java, SQL, Bash
Frameworks & Cloud: React, Next.js, FastAPI, Flask, Express, AWS (Lambda, DynamoDB, SAM), Supabase, Vercel
Databases & Tools: PostgreSQL, MongoDB, Git, Linux, Docker, pytest, CMocka, Playwright
AI & APIs: Claude, OpenAI and Gemini APIs, LiteLLM, Google ADK, Whisper, ElevenLabs
Spoken: Fluent in English, French, and Mandarin

Work Experience

Ciena Ottawa, ON
SWE & Tools Developer (returning co-op, extended part-time) *May 2026 – Present*

- Built a production PR review agent in Python on Claude Sonnet, run via Bitbucket webhooks on many repos
- Used daily by senior engineers, leads and managers; approves clean PRs and flags only critical/high-severity bugs
- Guarded against hallucinations: every finding must quote an exact diff line and cite symbols confirmed via grep
- Gave it grep and file-read tools on a cloned branch (10-call cap), plus pre-flight filters for known false positives
- Built a merge conflict resolver that resolves conflicts between two branches and raises a Jira-linked PR

Software Developer Intern *May 2025 – September 2025*

- Implemented Sherlog, a system logging facility in C/C++ and Python, unit tested with CMocka
- Built an agent explaining C memory leaks from LeakSanitizer and Heaptrack output; caught a real leak at Ciena
- Ran demos and tutorials on the leak agent for engineers

Projects

Ming | *Next.js, Claude API, Google Cloud TTS, Whisper, Supabase, Expo* June 2026 – Present

- Building an AI Mandarin coaching app with persona-based AI coaches and voice I/O, deployed on Vercel
- Kept LLM cost per message flat with server-side per-user daily caps, trimmed context, and capped replies
- Wrote a spaced-repetition scheduler unit tested in CI; ships as a PWA and an Expo iOS app on the same backend

Depth | *JavaScript, MediaPipe, Computer Vision* September 2026

- Built a browser squat game that scores every rep by depth from webcam pose tracking, all on-device
- Computed 3D knee angles with per-player calibration and smoothed pose jitter with a One Euro filter
- Counted reps with a debounced state machine; four game modes plus a no-camera demo player

Stash | *Python, AWS Lambda, DynamoDB, AWS SAM* October 2026

- Built a serverless link saver on one AWS Lambda and DynamoDB table, deployed with AWS SAM in the free tier
- Hardened link previews against SSRF; added CloudWatch alarms, SNS alerts, and a zero-spend budget alarm
- Tested with pytest and moto against mocked DynamoDB; added a Google ADK agent as a container Lambda

Pinky | *Next.js, TypeScript, Supabase, Postgres, Vercel* October 2026

- Built a live app where friend groups make accountability pacts, check in daily, and pay a shared pot on misses
- Enforced streaks, escalating stakes, and photo-proof challenges with Postgres functions and row-level security
- Deployed on Vercel with a daily reminder cron job and web push notifications

Leadership & Hackathons

McHacks: Experience Coordinator for McGill's hackathon (Sep 2026 – Present), running participant experience
Hackathons: Dishly at cuHacking 7 (Express, Gemini, Playwright tests); AI mock interviewer at McHacks 13